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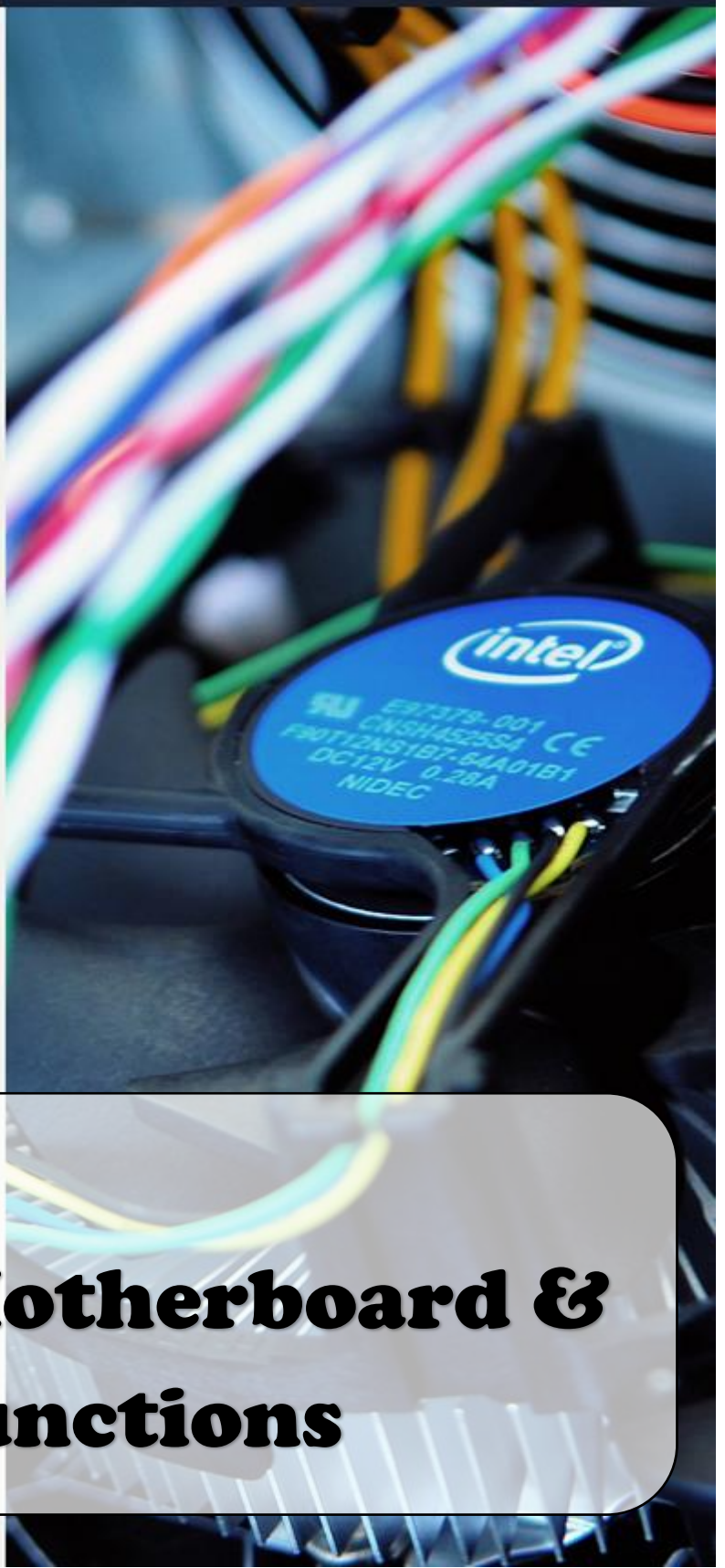
DEPARTMENT OF EDUCATION

COMPUTER SYSTEMS SERVICING NC II
GRADE 11

QUARTER 1
COC 1:
Install and Configure Computer Systems



Module 4:
Parts of the Motherboard & their Functions

Computer Systems Servicing NCII – Grade 11
Quarter 1 – Module 4: Parts of the Motherboard & their Functions
First Edition, 2020

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Introductory Message

For the facilitator:

Welcome to the **Computer Systems Servicing NCII G11** module on **Parts of the Motherboard & their Functions**.

This module was collaboratively designed, developed and reviewed by educators from Schools Division Office of Pasig City headed by its Officer-In-Charge Schools Division Superintendent, Ma. Evalou Concepcion A. Agustin in partnership with the Local Government of Pasig through its mayor, Honorable Vico Sotto.

The writers utilized the standards set by the K to 12 Curriculum using the Most Essential Learning Competencies (MELC) while overcoming their personal, social, and economic constraints in schooling.

This learning material hopes to engage the learners into guided and independent learning activities at their own pace and time. Further, this also aims to help learners acquire the needed 21st century skills especially the 5 Cs namely: Communication, Collaboration, Creativity, Critical Thinking and Character while taking into consideration their needs and circumstances.

In addition to the material in the main text, you will also see this box in the body of the module:

Notes to the Teacher

This contains helpful tips or strategies that will help you in guiding the learners.

As a facilitator you are expected to orient the learners on how to use this module. You also need to keep track of the learners' progress while allowing them to manage their own learning. Moreover, you are expected to encourage and assist the learners as they do the tasks included in the module.

For the learner:

Welcome to Computer Systems Servicing NCII Module on Parts of System Unit and their Functions. The hand is one of the most symbolized part of the human body. It is often used to depict skill, action and purpose. Through our hands we may learn, create and accomplish. Hence, the hand in this learning resource signifies that you as a learner is capable and empowered to successfully achieve the relevant competencies and skills at your own pace and time. Your academic success lies in your own hands!

This module was designed to provide you with fun and meaningful opportunities for guided and independent learning at your own pace and time. You will be enabled to process the contents of the learning material while being an active learner.

This module has the following parts and corresponding icons:

Expectation - These are what you will be able to know after completing the lessons in the module

Pre-test - This will measure your prior knowledge and the concepts to be mastered throughout the lesson.

Recap - This section will measure what learnings and skills that you understand from the previous lesson.

Lesson- This section will discuss the topic for this module.

Activities - This is a set of activities you will perform.

Wrap Up- This section summarizes the concepts and applications of the lessons.

Valuing-this part will check the integration of values in the learning competency.

Post-test - This will measure how much you have learned from the entire module. Ito po ang parts ng module

EXPECTATIONS

After completing this lesson, you should be able to:

1. identify the components of the motherboard;
2. understand functions of the components of the motherboard;
3. appreciate the importance of learning the components of motherboard.

PRETEST

DIRECTION: Read the following statement carefully. Choose the best answer by encircling the letter of your choice.

1. What is the portion in the motherboard which holds the CPU (Central Processing Unit) or simply as processor?
   a. CPU Holder c. Power Supply Controller
   b. RAM Holder d. North Bridge Chipset
2. It is the portion where you can connect the power supply unit (PSU) cable connector.
   a. CPU Holder c. Power Supply Controller
   b. RAM Holder d. North Bridge Chipset
3. This is the slot where you will insert the memory module or memory card.
   a. CPU Holder c. Power Supply Controller
   b. RAM Holder d. North Bridge Chipset
4. This handles the data-transfer duties of memory, CPU, and AGP and to make the most efficient use of available resources.
   a. CPU Holder c. Power Supply Controller
   b. RAM Holder d. North Bridge Chipset
5. This component provides support for a wide variety of devices with many differing bus speeds and designs.
   a. IDE Controller c. AGP
   b. BIOS d. South Bridge Chipset
6. It is the built-in software that tells what a computer can do without accessing programs from a disk.
   a. IDE Controller c. AGP
   b. BIOS d. South Bridge Chipset
7. This connector is used to insert an IDE cable.
   a. IDE Slot c. AGP
   b. BIOS d. South Bridge Chipset
8. It was designed specifically for AGP video cards.
   a. IDE Controller c. AGP
   b. BIOS d. South Bridge Chipset

9. These are pins in the motherboard that will connect the Power Button, Reset Button, Power LED, Mini Speaker, USB ports and other ports to the motherboard.

a. PCI-E Slot c. Front Panel Pins
b. BIOS d. South Bridge Chipset

10. It was designed to replace PCI, PCI-X, and AGP.

a. PCI-E Slot c. Front Panel Pins
b. BIOS d. South Bridge Chipset

RECAP

DIRECTION: Identity the following components of System Unit. Write your answer on the space provided before the number.

1. _____

2. _____

3. _____

4. _____

5. _____

LESSON

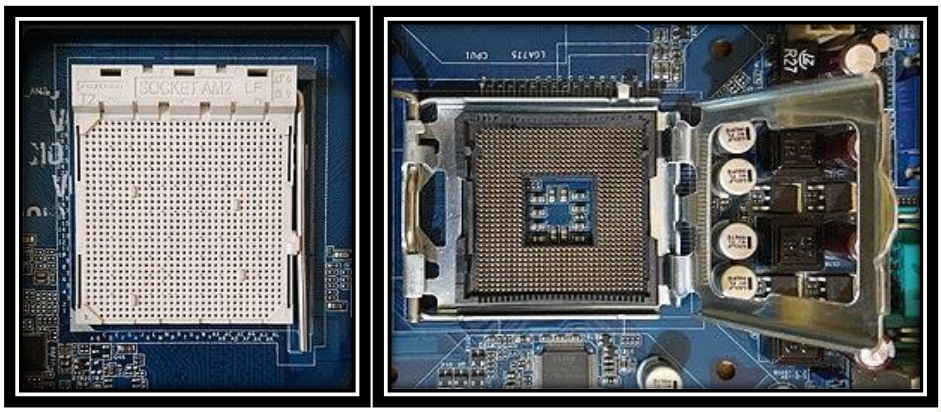
PARTS OF MOTHERBOARD & THEIR FUNCTIONS

The motherboard is one of the components of the system unit that performs major functions to run our computer. All the devices (input & output) we use and other important parts of our computer are attached to our motherboard.

In this lesson, we are going to learn the parts of the Motherboard and the functions of every component.

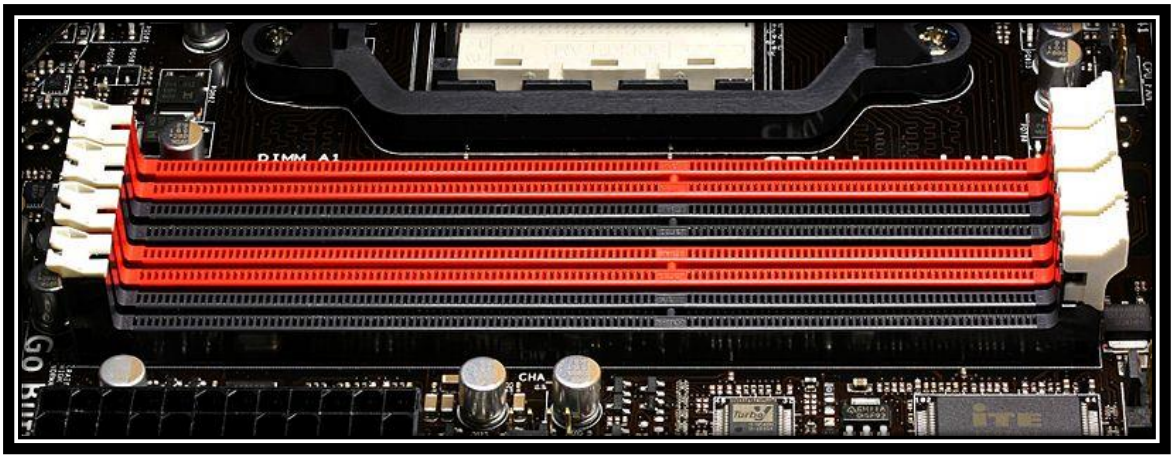
1. CPU Socket

- Also known as CPU Slot or CPU Holder
- This component of the motherboard holds the CPU (Central Processing Unit) and allows replacing of the component without soldering.



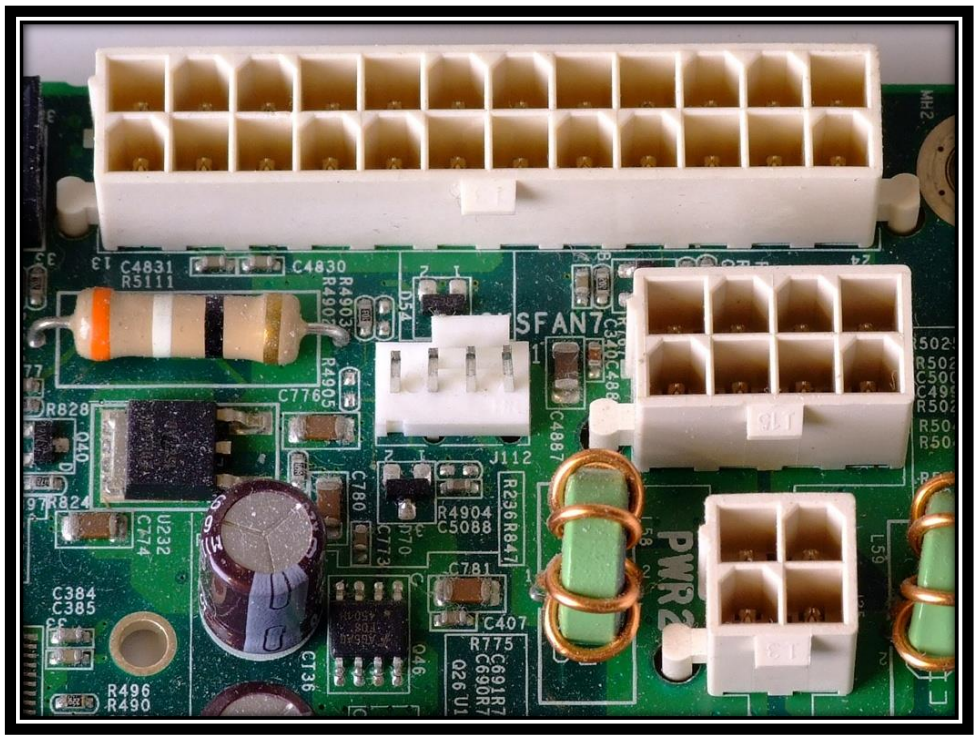
2. RAM Holder

- Also known as Memory Holder or Memory Slot
- This component is where the memory module or memory card will be inserted.
- Some motherboard comes with two to four slots.



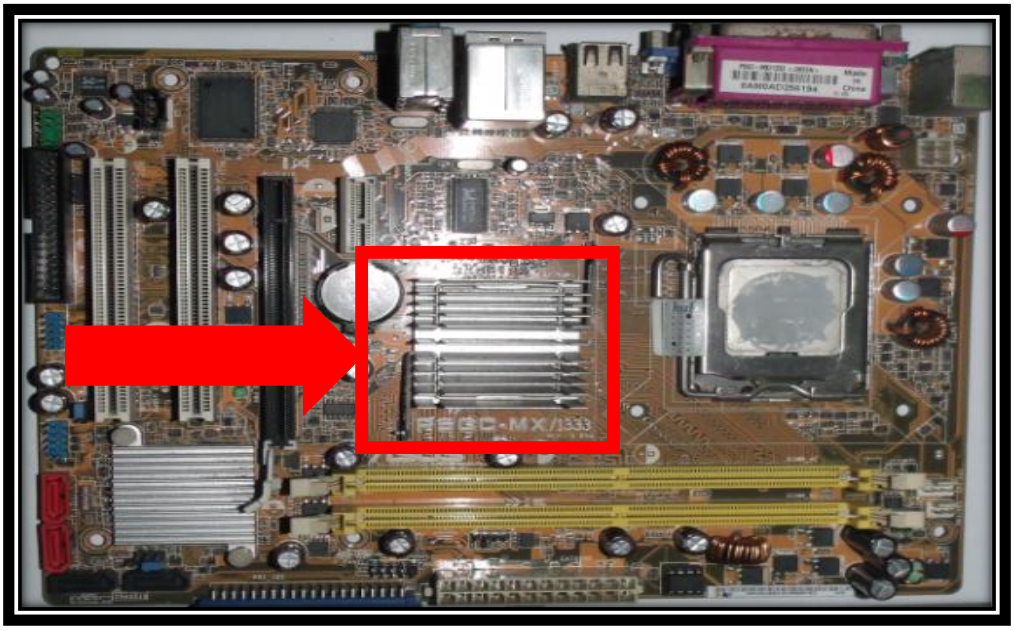
3. Power Supply Connector

- This part of the motherboard is where the power supply unit cable connector can be connected
- Types of PSU Connector
    1. 24 pins – main power connector that provides power to the motherboard
    2. 4 pins – provides power to the CPU
    3. 8 pins – provides power for faster dedicated graphics cards



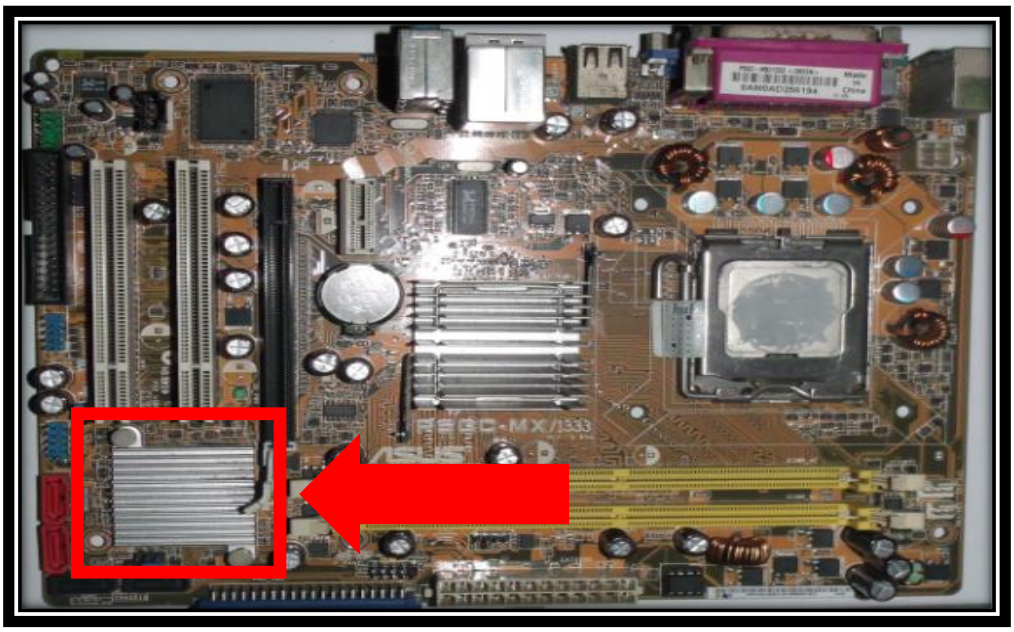
4. Northbridge Chipset

- Also known as Memory Controller Hub
- This chipset handles the data-transfer of memory, CPU, and Video Card thus responsible for tasks that require the highest performance.



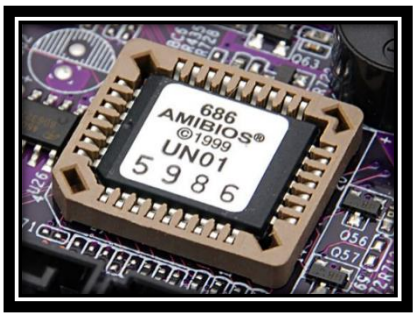
5. Southbridge Chipset

- This chipset component provides support for a wide variety of devices with many differing bus speeds and designs.
- Control over secondary buses such as USB, IDE, PS/2, BIOs and Ethernet.



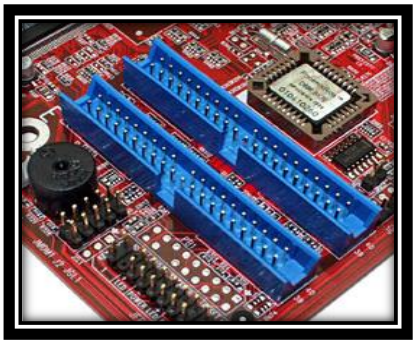
6. CMOS or BIOS Chipset

- (Complementary Metal Oxide Semi-conductor) is also a chipset which contains the BIOS (Basic Input Output System).



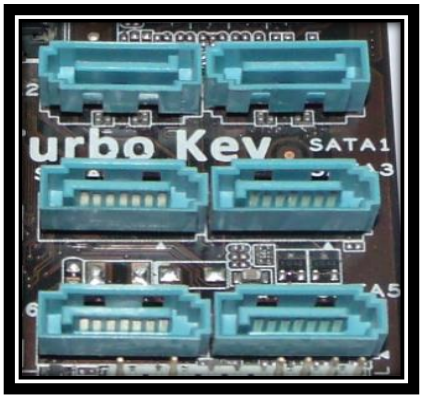
7. EIDE Slot/Port

- This part of the motherboard is where the IDE cables are connected intended for drives including Hard Disk Drive (HDD) and Optical Disk Drive (ODD).



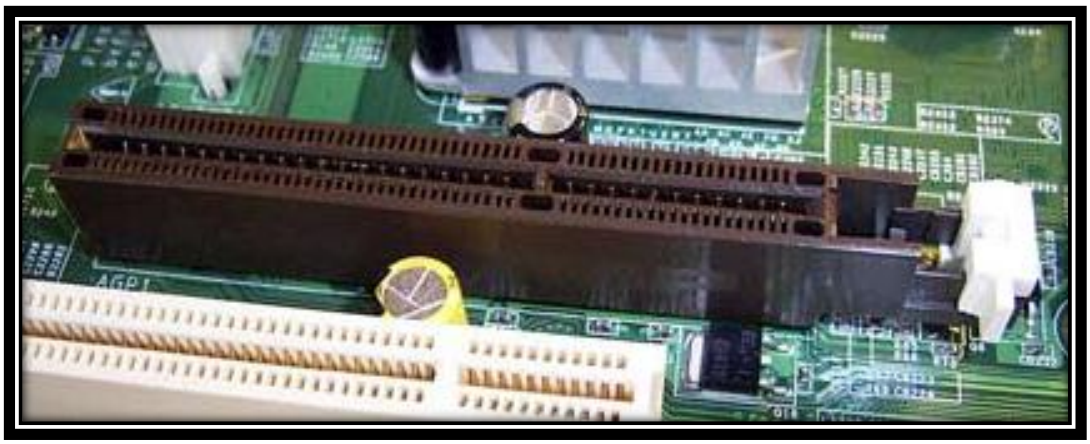
8. SATA Port/Slot

- This is used to connect SATA drives into the motherboard including Hard Disk Drive (HDD) and Optical Disk Drive (ODD).
- SATA (Serial ATA) has a faster transfer rate than IDE



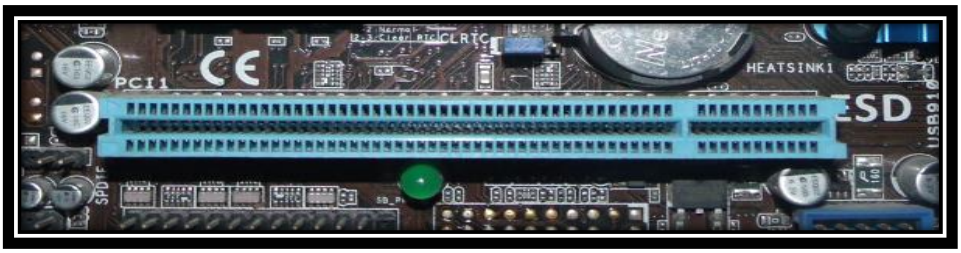
9. AGP (Accelerated Graphic Port)

- It was designed specifically for AGP video cards.
- It was originally designed as a successor to PCI-type connections for video cards.



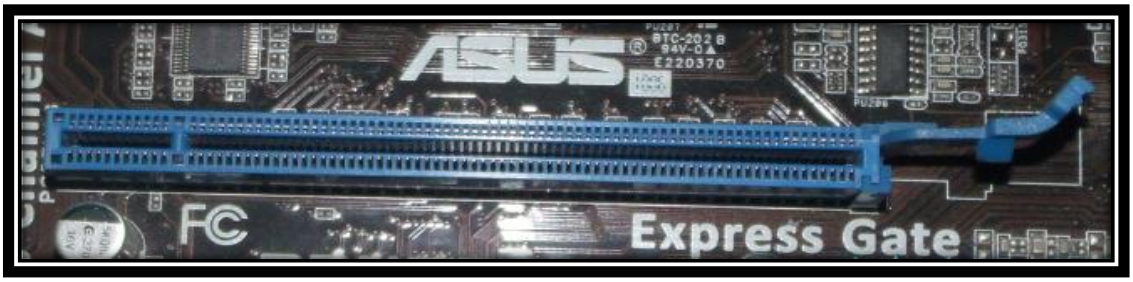
10. PCI (Peripheral Component Interconnect) Slot

- It is a built-in slot on a motherboard that allows for the attachment of various hardware components such as network cards, modems, sound cards, disk controllers and other peripherals.



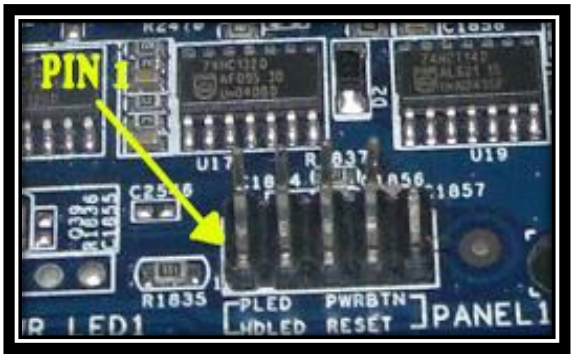
11. PCI-EXPRESS Slot

- PCIe was designed to replace PCI, PCI-X, and AGP.
- It is the most recent and high-performance standard for expansion cards that is generally available on modern personal computers.



11. Front Panel Pins

- These are pins in the motherboard that will connect the Power Button, Reset Button, Power LED, Mini Speaker, USB ports and other ports to the motherboard



12. PS2 port

- It is used to connect PS2 mouse and keyboard

13. USB Port

- This part of the motherboard is used to connected USB Devices and peripherals

14. HDMI Port

- It is used to connect HDMI cable from the monitor to the motherboard

15. Active Display port

- It is used to connect Active Display cable from the monitor to the motherboard

16. VGA Port

- It is used to connect VGA cable from the monitor to the motherboard

17. DVI Port

- It is used to connect DVI cable from the monitor to the motherboard

18. IEEE 1394a Port

- It is an interface standard for a serial bus for high-speed communications and isochronous real-time data transfer.
- It was developed in the late 1980s and early 1990s by Apple, which called it FireWire, in cooperation with a number of companies, primarily Sony and Panasonic.

19. E-SATA Port

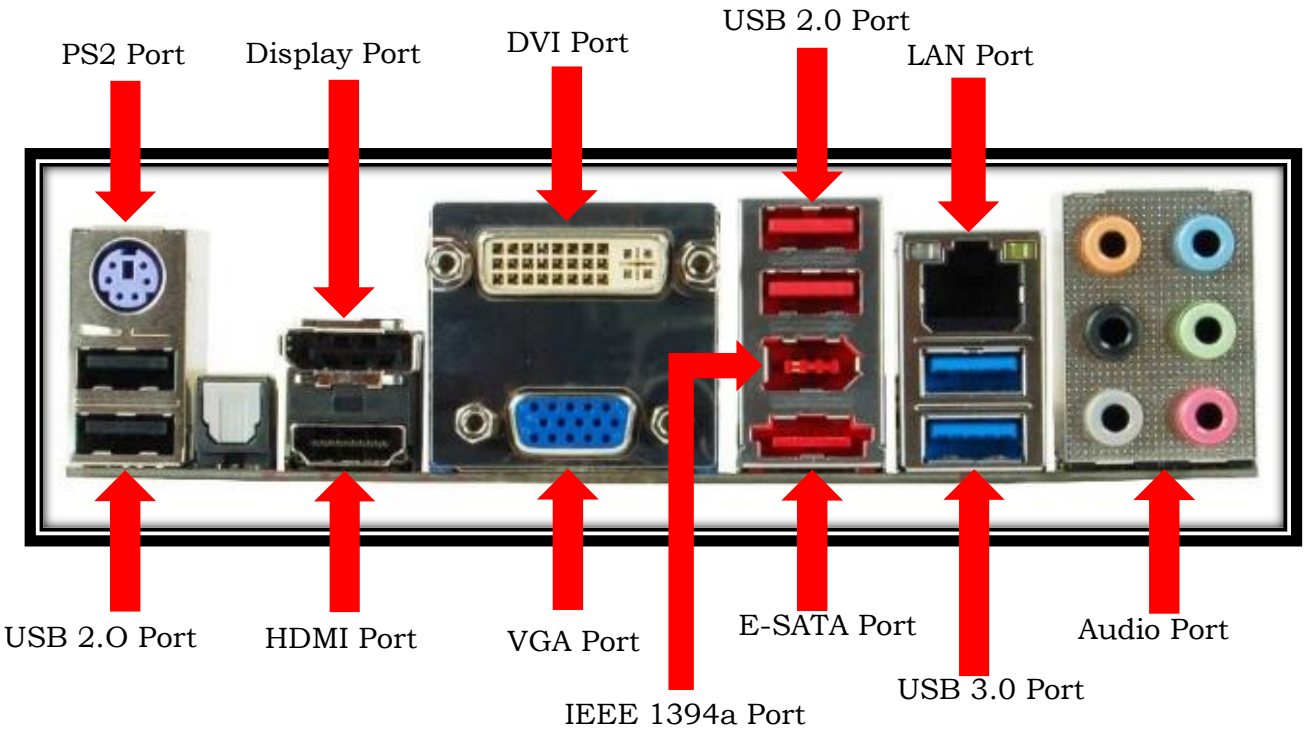
- It is a SATA connector accessible from outside the computer, to provide a signal (but not power) connection for external storage devices.

20. LAN Port

- Also known as Network Port
- LAN port allows a computer to connect to a network using a wired connection.

21. Audio Port

- An audio port is used for an audio device such as speakers, headphones or a microphone to be connected.



ACTIVITIES

Activity # 1: IDENTIFICATION

Direction: Identify what is being asked in the following statements. Write your answer on the space provided.

_____1. It is used to connect ActiveDisplay cable from the monitor to the motherboard.

_____2. This port is used for an audio device such as speakers, headphones or a microphone to be connected.

_____3. This is used to connect SATA drives into the motherboard including Hard Disk Drive (HDD) and Optical Disk Drive (ODD).

_____4. This component of the motherboard holds the CPU (Central Processing Unit) and allows replacing of the component without soldering.

_____5. This component is where the memory module or memory card will be inserted.

_____6. This chipset handles the data-transfer of memory, CPU, and Video Card thus responsible for tasks that require the highest performance.

_____7. It is also a chipset which contains the BIOS (Basic Input Output System).

_____8. It was originally designed as a successor to PCI-type connections for video cards.

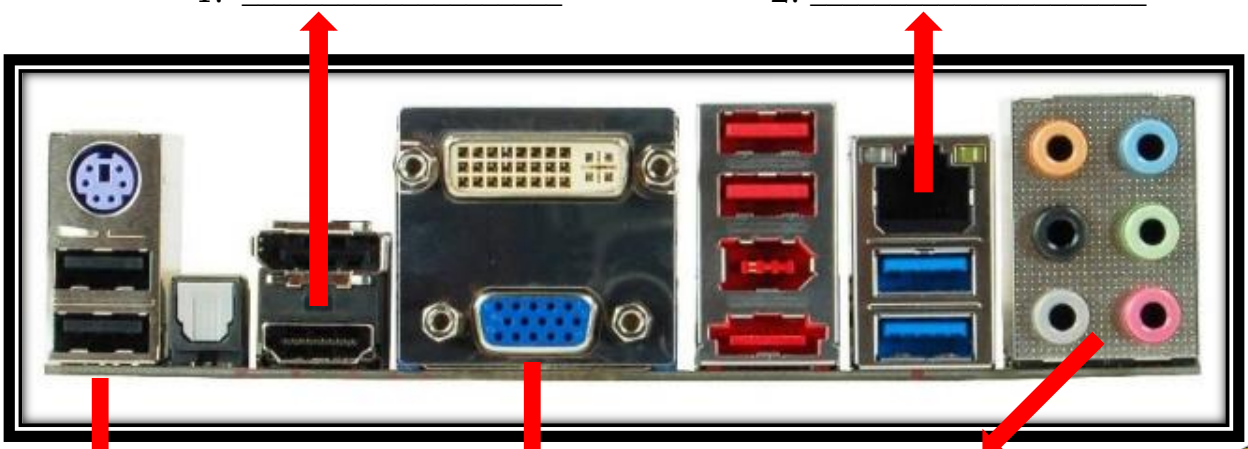
_____9. It is the most recent and high-performance standard for expansion cards that is generally available on modern personal computers.

_____10. These are pins in the motherboard that will connect the Power Button, Reset Button, Power LED, Mini Speaker, USB ports and other ports to the motherboard.

Activity # 2: LABEL ME

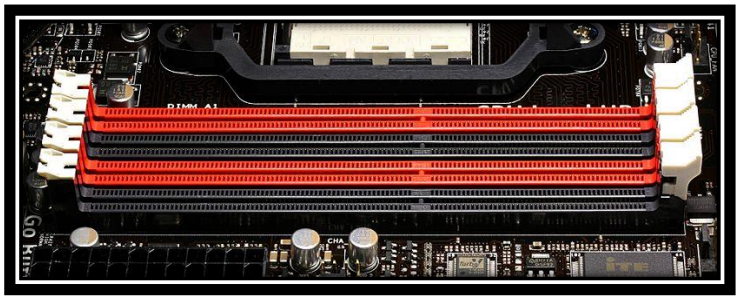
Direction: Label the following components of the Motherboard. Write your answer on the space provided before the number.

1. _____ 2. _____

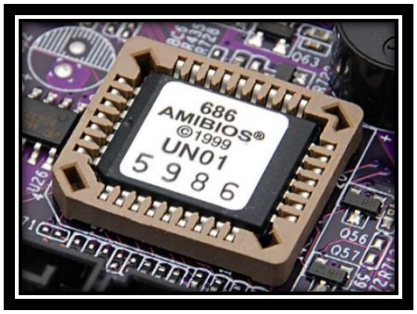


3. _____ 4. _____ 5. _____

_____6.



_____7.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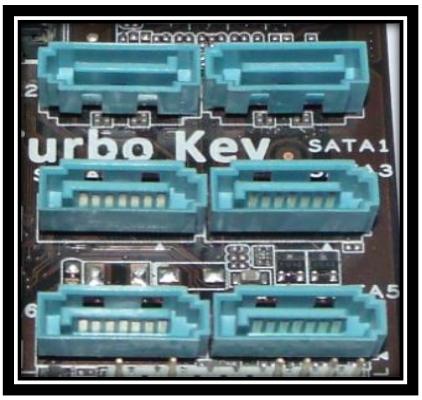
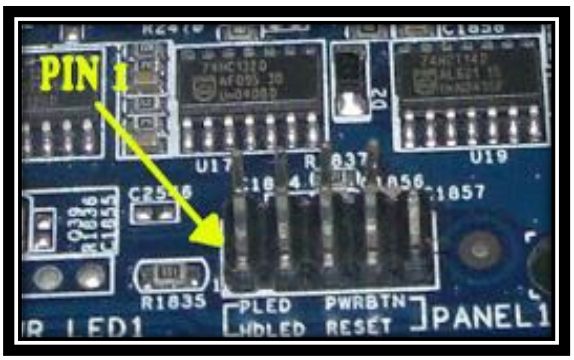
_____8.


urbo Key
SATA1

_____9.


PIN 1
U17
U19
R1835
PLED
PWRBTN
HDLED
RESET
PANEL1

_____10.

WRAP–UP

DIRECTION: Complete the following sentences base on what you've learn.

1. The components of the motherboard are as follows _____
_____
_____
_____
_____.

2. Write the function/s of the following parts of motherboard:
   a. CPU Slot_____
   b. RAM Slot _____
   c. VGA Port _____
   d. SATA Port _____
   e. HDMI Port _____

VALUING

DIRECTION: Read the following questions carefully. State your answer in two to three sentences each number on the space provided.

1. What do you think are the importance of learning the components of motherboard?
_____
_____
_____
_____

2. How will you use the knowledge you acquired about components of Motherboard?
_____
_____
_____
_____

POST TEST

DIRECTION: Read the following statement carefully. Choose the best answer by encircling the letter of your choice.

1. It was designed to replace PCI, PCI-X, and AGP.
   a. PCI-E Slot c. Front Panel Pins
   b. BIOS d. South Bridge Chipset
2. What is the portion in the motherboard which holds the CPU (Central Processing Unit) or simply as processor?
   a. CPU Holder c. Power Supply Controller
   b. RAM Holder d. North Bridge Chipset
3. This handles the data-transfer duties of memory, CPU, and AGP and to make the most efficient use of available resources.
   a. CPU Holder c. Power Supply Controller
   b. RAM Holder d. North Bridge Chipset
4. It is the portion where you can connect the power supply unit (PSU) cable connector.
   a. CPU Holder c. Power Supply Controller
   b. RAM Holder d. North Bridge Chipset
5. This is the slot where you will insert the memory module or memory card.
   a. CPU Holder c. Power Supply Controller
   b. RAM Holder d. North Bridge Chipset
6. This component provides support for a wide variety of devices with many differing bus speeds and designs.
   a. IDE Controller c. AGP
   b. BIOS d. South Bridge Chipset
7. It is the built-in software that tells what a computer can do without accessing programs from a disk.
   a. IDE Controller c. AGP
   b. BIOS d. South Bridge Chipset
8. This connector is used to insert an IDE cable.
   a. IDE Slot c. AGP
   b. BIOS d. South Bridge Chipset
9. It was designed specifically for AGP video cards.
   a. IDE Controller c. AGP
   b. BIOS d. South Bridge Chipset
10. These are pins in the motherboard that will connect the Power Button, Reset Button, Power LED, Mini Speaker, USB ports and other ports to the motherboard.
   a. PCI-E Slot c. Front Panel Pins
   b. BIOS d. South Bridge Chipset

KEY TO CORRECTION

Pretest	REGAP	Activity # 1:
1. a 2. c 3. b 4. d 5. d 6. b 7. a 8. c 9. c 10.a	1. Video Card/Graphic Card 2. Optical Disk Drive 3. Hard Disk Drive 4. RAM 5. Motherboard	1. ActiveDisplay Port 2. Audio Port 3. SATA Port 4. CPU Slot/Socket 5. RAM Holder / RAM Slot 6. Northbridge Chipset 7. CMOS 8. AGP 9. PCI-e Slot 10. Front Panel Pins
Activity # 2: 1. SATA Port 2. LAN Port 3. USB Port 4. VGA Port 5. Audio Port 6. RAM Slot 7. CMOS 8. SATA Port 9. Front Panel Pins 10.DVI Port		Post Test 1. a 2. a 3. d 4. c 5. b 6. d 7. b 8. a 9. c 10.c

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